

Getting your Python running

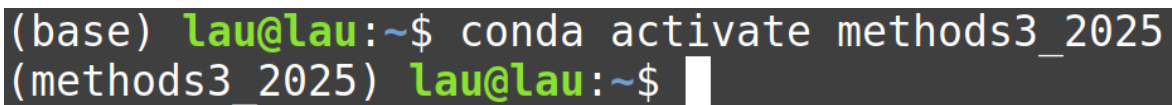
IDE (Integrated Development Editor) options for running Python

- RStudio - the easy, but maybe unorthodox choice
 - You will be able to knit Rmd files with Python code in
- VSCode - the obvious choice, many of you are running it already
 - can convert notebooks, *.ipynb*, to pdf's, similar to knitting
- Spyder - the old-fashioned steady choice
 - you can add notebook-support

Choice for next year: setting it up on UCloud!!

Prerequisites: Having installed the *methods3_2025* environment (see the [Preamble](#))

How to check: from a terminal (Linux/Mac) or Anaconda Powershell Prompt (Windows)
run `conda activate methods3_2025`



```
(base) lau@lau:~$ conda activate methods3_2025
(methods3_2025) lau@lau:~$
```

Figure 1: Activate conda env

RStudio

Install the package *reticulate*: `install.packages('reticulate')`

Running `conda_list()` should now reveal all your environments

Running `use_condaenv("methods3_2025", required = TRUE)` changes the environment to *methods3_2025*

Check that it works:

```
import sklearn
```

Troubleshooting

If you get an error looking like this after `usecondaenv`:

```
> use_condaenv('methods3_2025', required=TRUE)
```

```
Warning in system2(command = python, args = shQuote(script), stdout = TRUE, :  
  kørende kommando ''/home/lau/miniconda3/envs/methods3_2025/bin/python' '/usr/lib/R/site-library/reticu  
late/config/config.py' 2>/dev/null' havde status 1
```

then exchange `config.py` at the path, (in my case): `'/usr/lib/R/site-library/reticulate/config/config.py'` with the following `config.py` file [Config File](#)

This will require administrative privileges

VSCode

I have the least experience here, but I have tested the suggestions from Lumo (a GPT):

2 Install the Python extension (if you haven't already)

1. Press **Ctrl + Shift + X** (or click the square-shaped **Extensions** icon on the left sidebar).
2. In the search box type **"Python"**.
3. Choose **"Python"** published by **Microsoft** and click **Install**.
4. Reload VS Code when prompted.

The extension supplies the interpreter picker, linting, debugging, Jupyter support, etc.

Figure 2: Install python extension

4 Tell VS Code which Conda interpreter to use

1. Open a folder or workspace in VS Code (`File → Open Folder...`).
2. Press **Ctrl + Shift + P** (or `⇧⌘P`) to open the **Command Palette**.
3. Type **"Python: Select Interpreter"** and hit **Enter**.
4. You'll see a list that includes entries like:

```
Conda (myenv) • C:\Users\<you>\miniconda3\envs\myenv\python.exe
```

5. Click the entry that corresponds to the environment you want.

VS Code writes the selection to a hidden settings file inside the workspace:

Figure 3: Find conda env

Then you can create notebooks: `.ipynb` which can be converted to pdf's, using `nbconvert` to do the equivalent of knitting, e.g. `jupyter nbconvert --to html --execute --output report.html analysis.ipynb`

Also check the video [VScode and conda](#)

Troubleshooting

Conda may not be on your `$PATH`

Windows

Opening Visual Studio Code from the Windows terminal

You can launch VS Code directly from any command-line session (Command Prompt, PowerShell, Windows Terminal, Git Bash, etc.) by using the `code` command. Below is a quick guide to get it working and a few handy shortcuts.

1 Make sure the `code` command is on your PATH

When you install VS Code you're asked whether to add the "`code`" command to your system `PATH`.

- If you checked that box during installation – you're ready to go.
- If you missed it – you can add it now:

Method	Steps
Via VS Code UI	1. Open VS Code. 2. Open the Command Palette (<code>Ctrl+Shift+P</code> / <code>⇧⌘P</code>). 3. Type " Shell Command: Install 'code' command in PATH " and press Enter . 4. Restart any open terminals.

Figure 4: Adding the path

Mac

Run `code` from a terminal after having put in `conda activate methods3_2025`, i.e. run `code`

VS Code launch	If you start VS Code from Finder/Dock , it inherits the PATH that macOS gives GUI apps (often a very minimal PATH). If you start VS Code from a terminal (<code>code .</code>), it inherits the terminal's PATH.
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Spyder

The best idea here is to run Spyder in its own environment

```
conda create -c conda-forge -n spyder-env spyder
conda activate spyder-env
spyder
```

When Spyder has opened, go into Console -> New console in environment -> pick the `methods_2025` one. It may ask you to install `spyder-kernels`

Open a new terminal

```
conda activate methods3_2025
conda install spyder-kernels=*.*. * ## the version that it asks for in the console
```

Then start a new console. Should be fixed now

Adding notebook support:

```
conda activate spyder-env
conda install spyder-notebook
```